

EE/CE 252/251

Signals and Systems

with

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*In the last few months, we
have seen...*

What is a **signal** and what is a **system**? (from engineering mathematics perspective)

Chapter 1

What are some of the common types of signals and systems?

Chapter 1

What are some of the **useful properties** of signals and systems?

Chapter 1

What are some of the major ways of
modelling and analyzing signals and systems?

Time
Domain
Laplace

Z-Transform

Fourier

Chapters

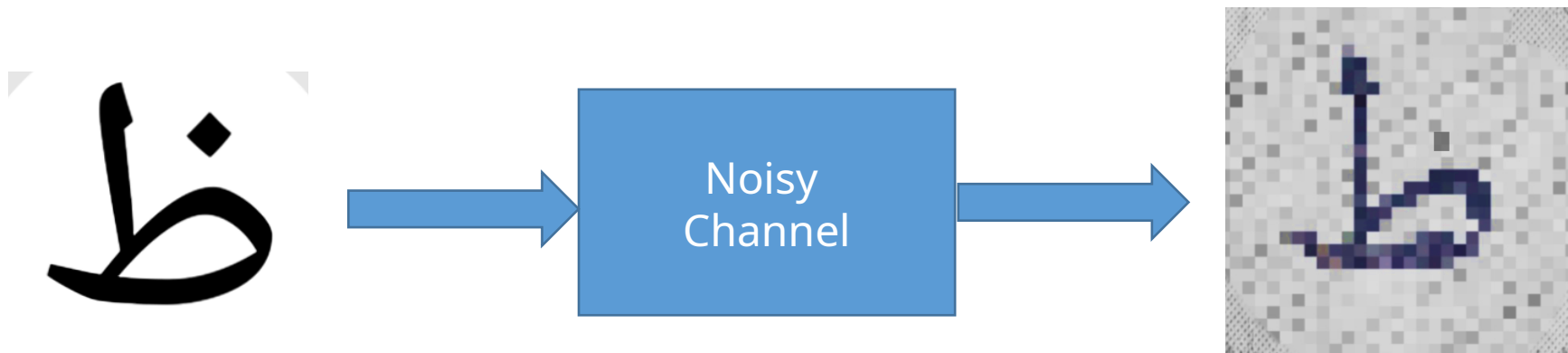
2, 3
Chapter 4

Chapter 5

Chapters 6, 7

How do signals and systems **interact**?

**Chapters 2-
7**



Let us now look at some practical applications of what we've learned...

- **Examples from my own research:** Material Identification
- **Analyzing circuits :** Laplace Transform
- **Wireless Communications :** Z Transform
- **Filter Design :** Fourier Transform

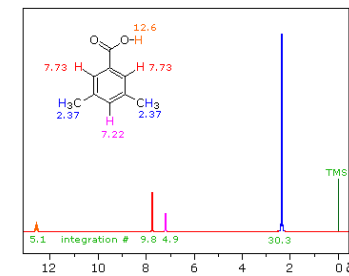
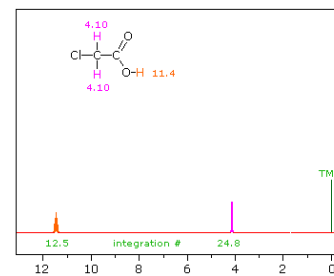
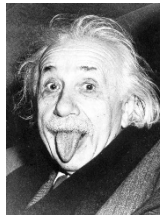
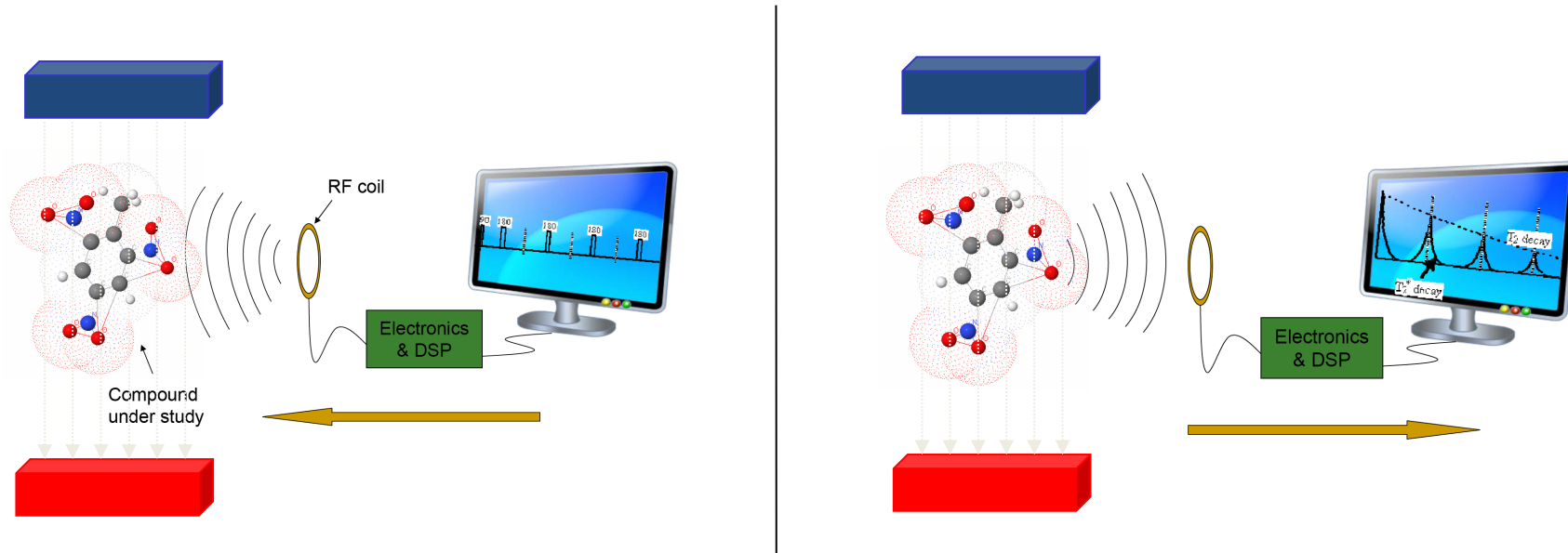
CONPHIRMER Project



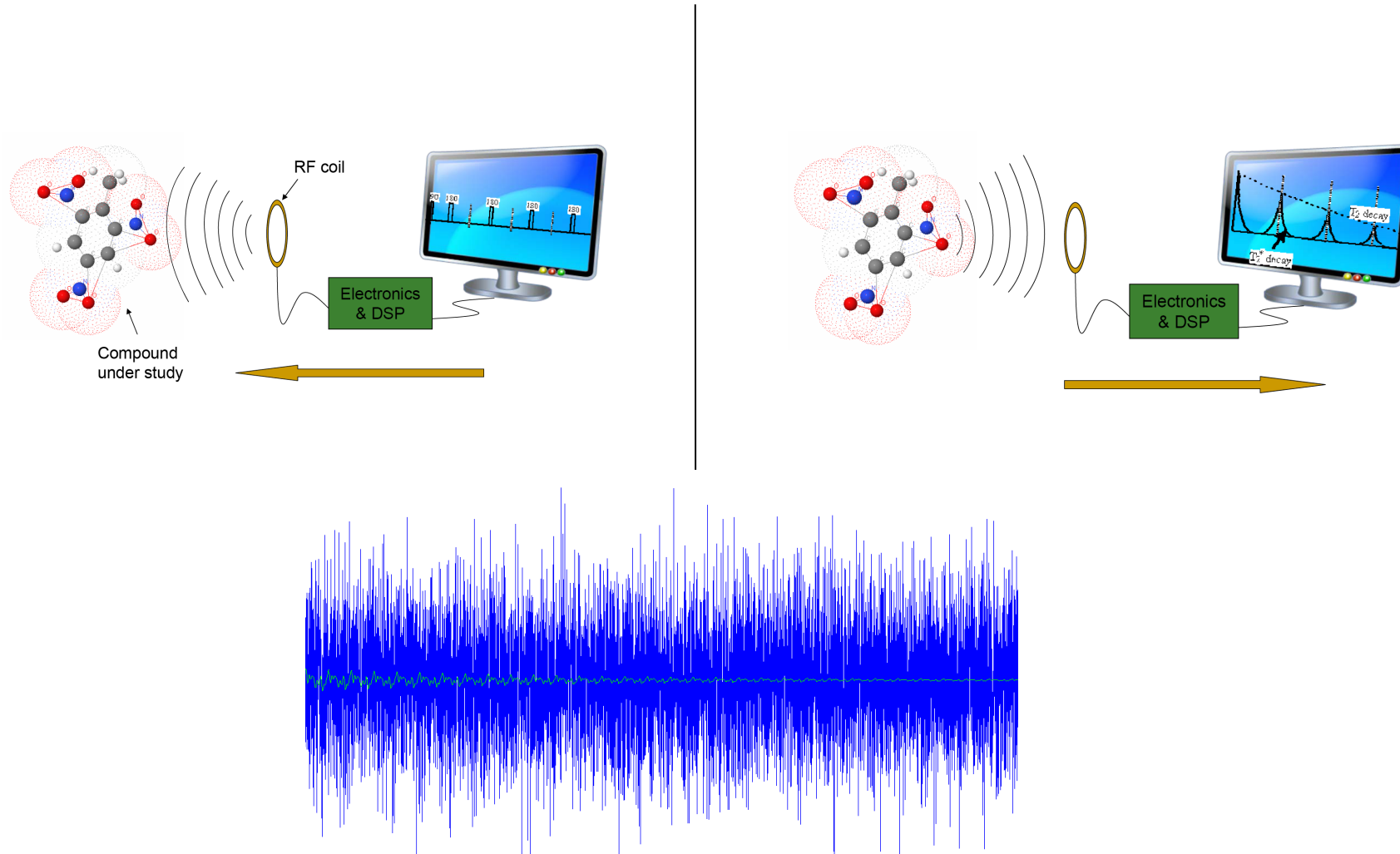
Question: how to quickly tell whether a medicine is fake?



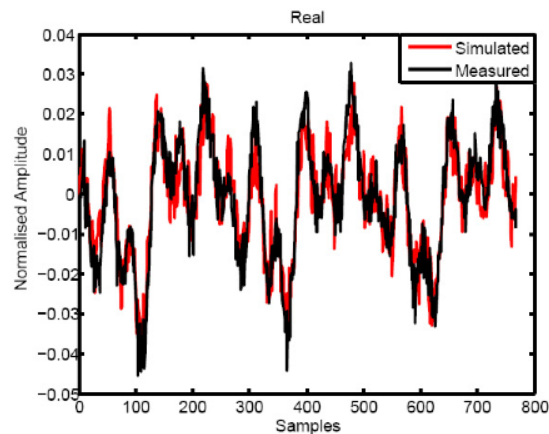
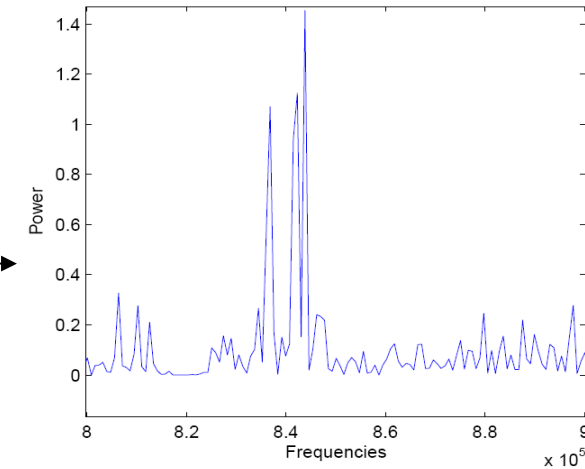
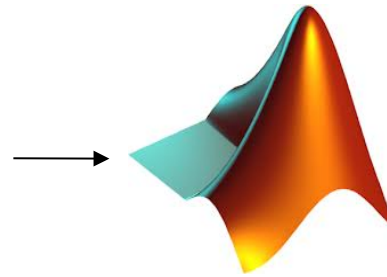
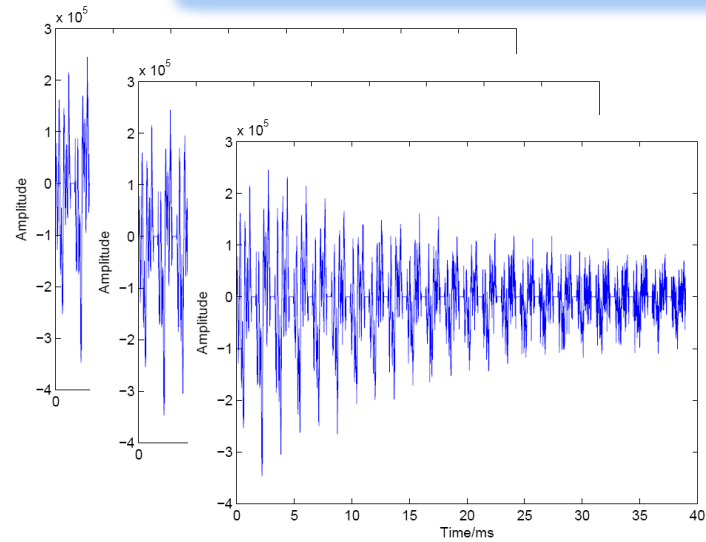
NMR vs. NQR



NQR Signal



NQR Signal Modeling



$$y_m^{(p)}(t) = \sum_{k=1}^{d^{(p)}} \alpha_k^{(p)} e^{-(t+m\mu)\eta_k^{(p)}} e^{-\beta_k^{(p)}|t-t_{sp}| + i\omega_k^{(p)}(T)t}$$

$$\omega_k^{(p)}(T) = a_k^{(p)} - b_k^{(p)}T$$

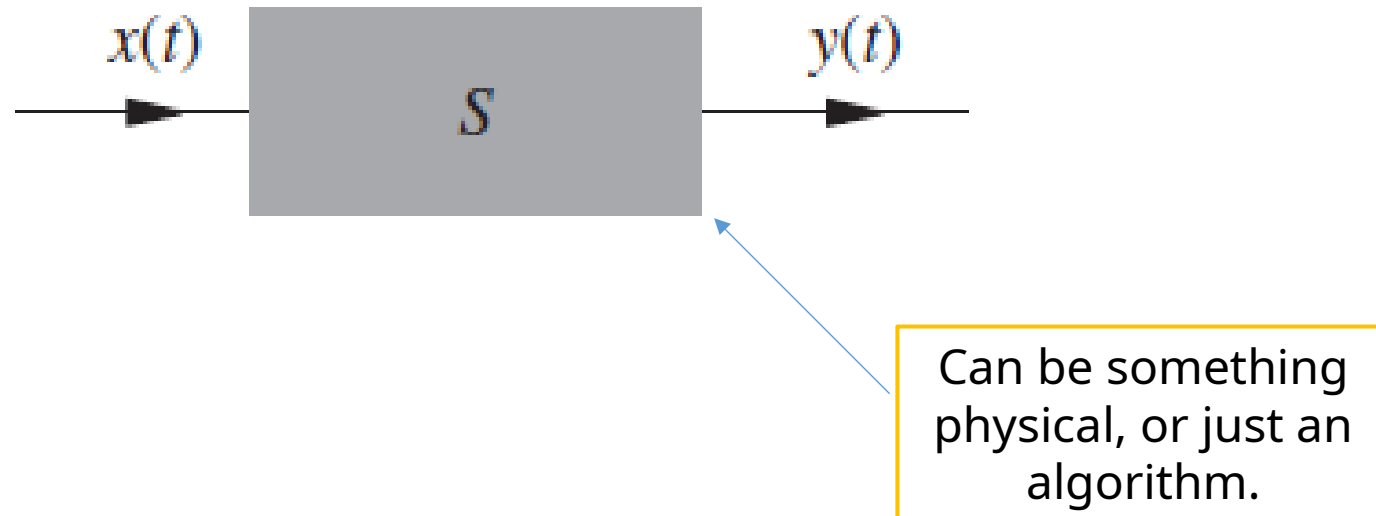
$$\alpha_k^{(p)} = \rho \kappa_k^{(p)} \quad \left\| \kappa - \kappa_a \right\|_2^2 \leq \epsilon$$

Let us now look at some practical applications of what we've learned...

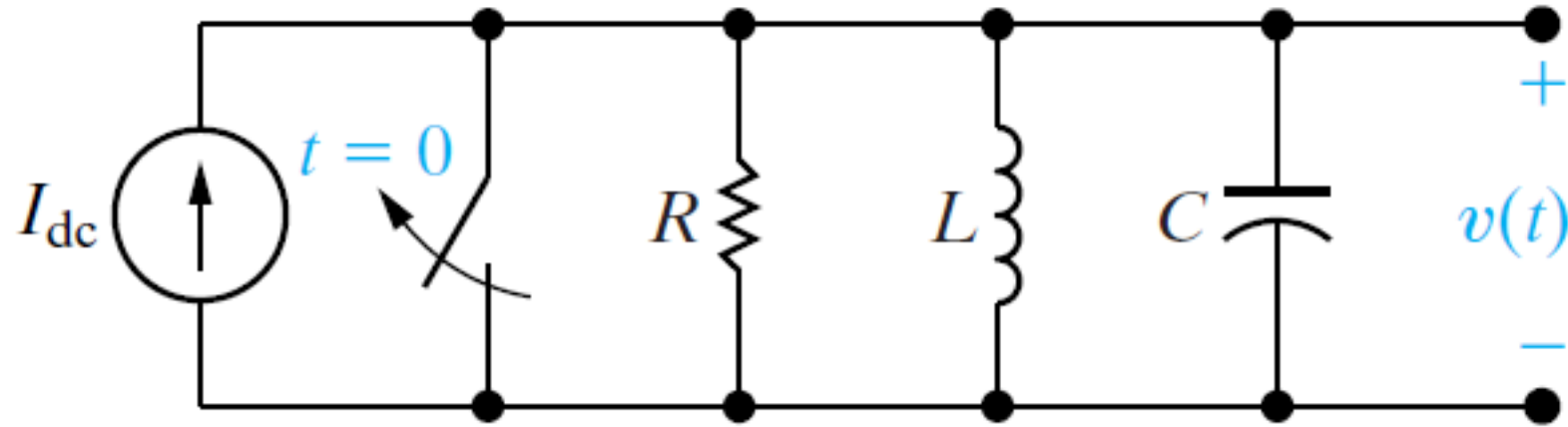
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"System"

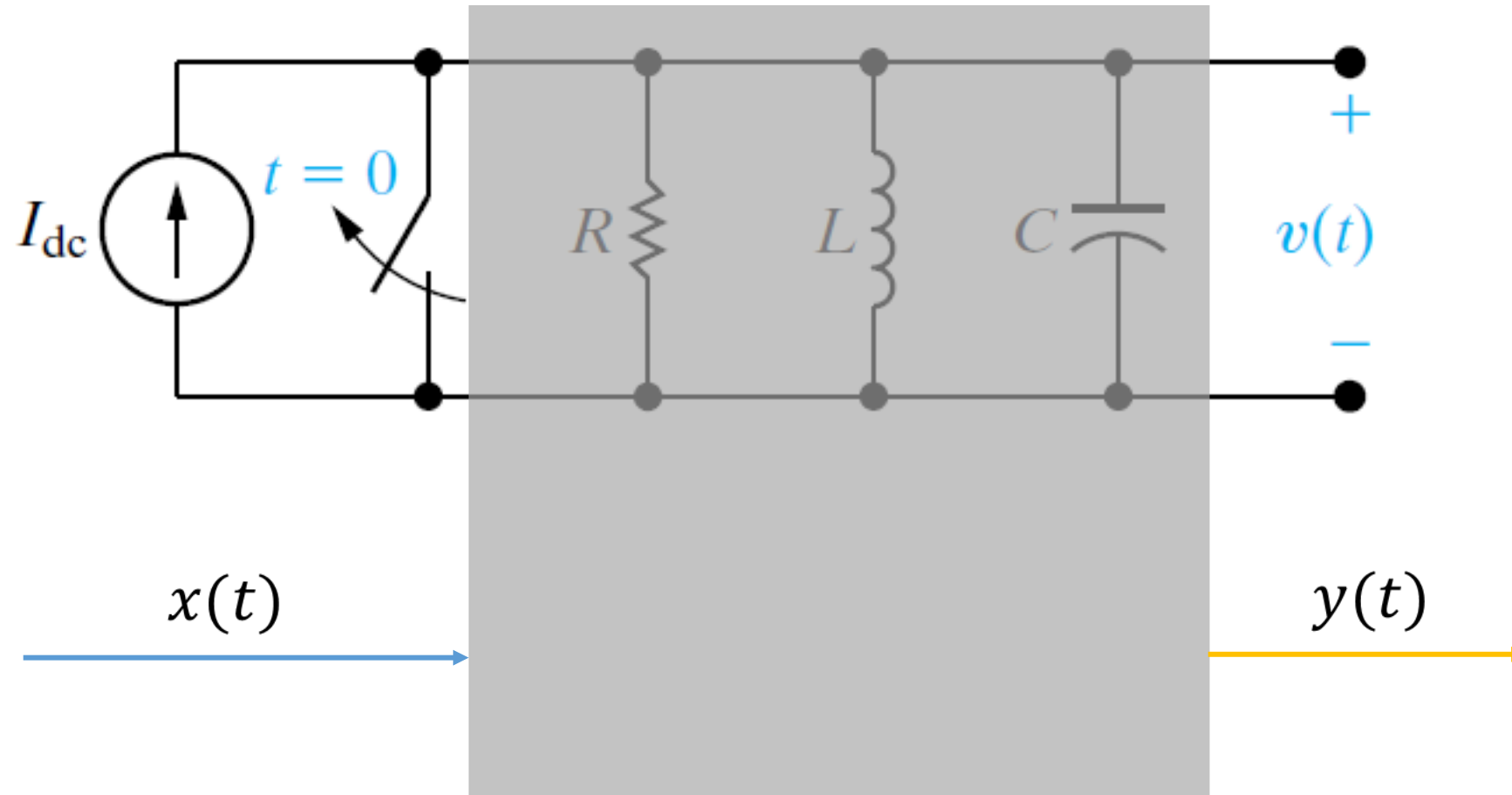
An entity that processes a signal



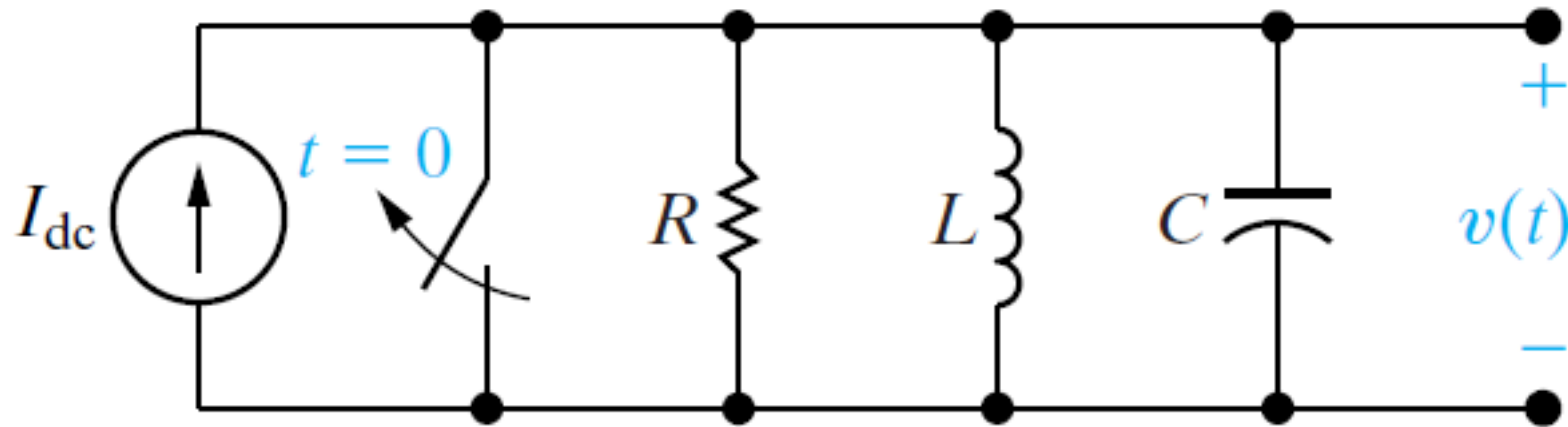
Example of a System



Example of a System



System Modeled as a Differential Equation (Time-Domain)



$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{dc} u(t).$$

Using Laplace (Frequency Domain) to Solve and Analyze the System

$$\frac{v(t)}{R} + \frac{1}{L} \int_0^t v(x) dx + C \frac{dv(t)}{dt} = I_{\text{dc}} u(t).$$

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{\text{dc}} \left(\frac{1}{s} \right),$$

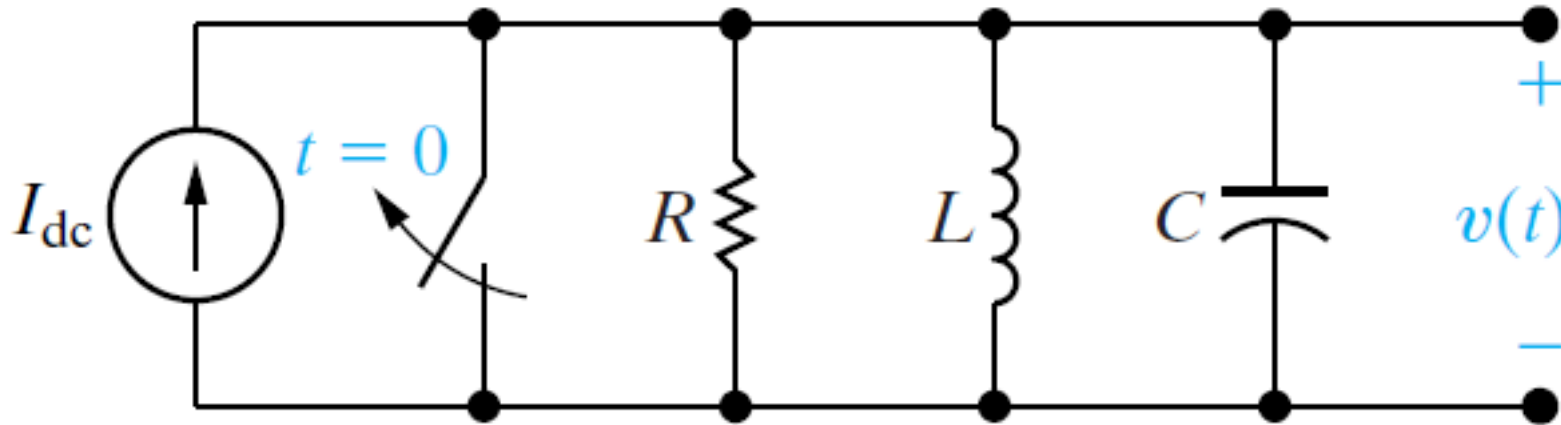
Using Laplace (Frequency Domain) to Solve and Analyze the System

$$\frac{V(s)}{R} + \frac{1}{L} \frac{V(s)}{s} + C[sV(s) - v(0^-)] = I_{dc} \left(\frac{1}{s} \right),$$

$$V(s) = \frac{I_{dc}/C}{s^2 + (1/RC)s + (1/LC)}$$

Put values and use PFE to get inverse Laplace.

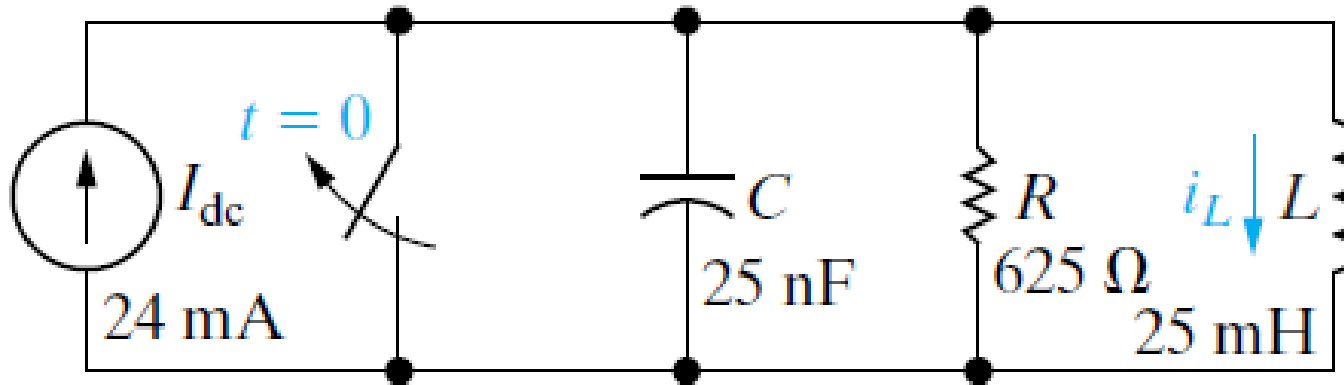
Using Laplace (Frequency Domain) to Solve and Analyze the System



$$v(t) = \mathcal{L}^{-1}\{V(s)\}$$

Analysis: what are the initial and final (steady-state) values of $i_L(t)$?

Assuming zero ICs



$$I_L = \frac{I_{dc}/LC}{s[s^2 + (1/RC)s + (1/LC)]}$$

$$i_L(\infty) = \lim_{s \rightarrow 0} sI_L(s) = I_{dc}$$

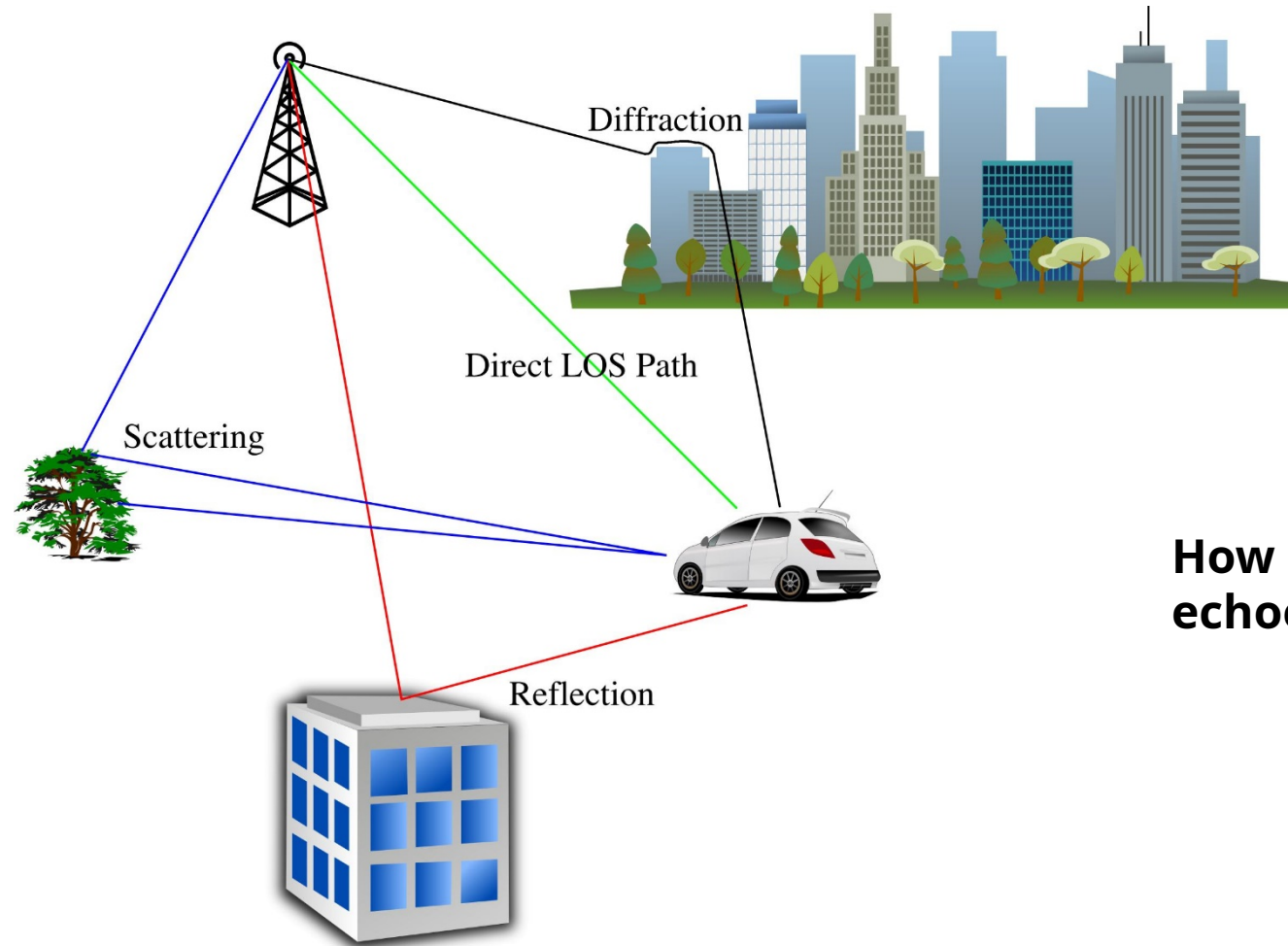
$$i_L(0) = \lim_{s \rightarrow \infty} sI_L(s) = 0$$

Do these values make sense?

Let us now look at some practical applications of what we've learned...

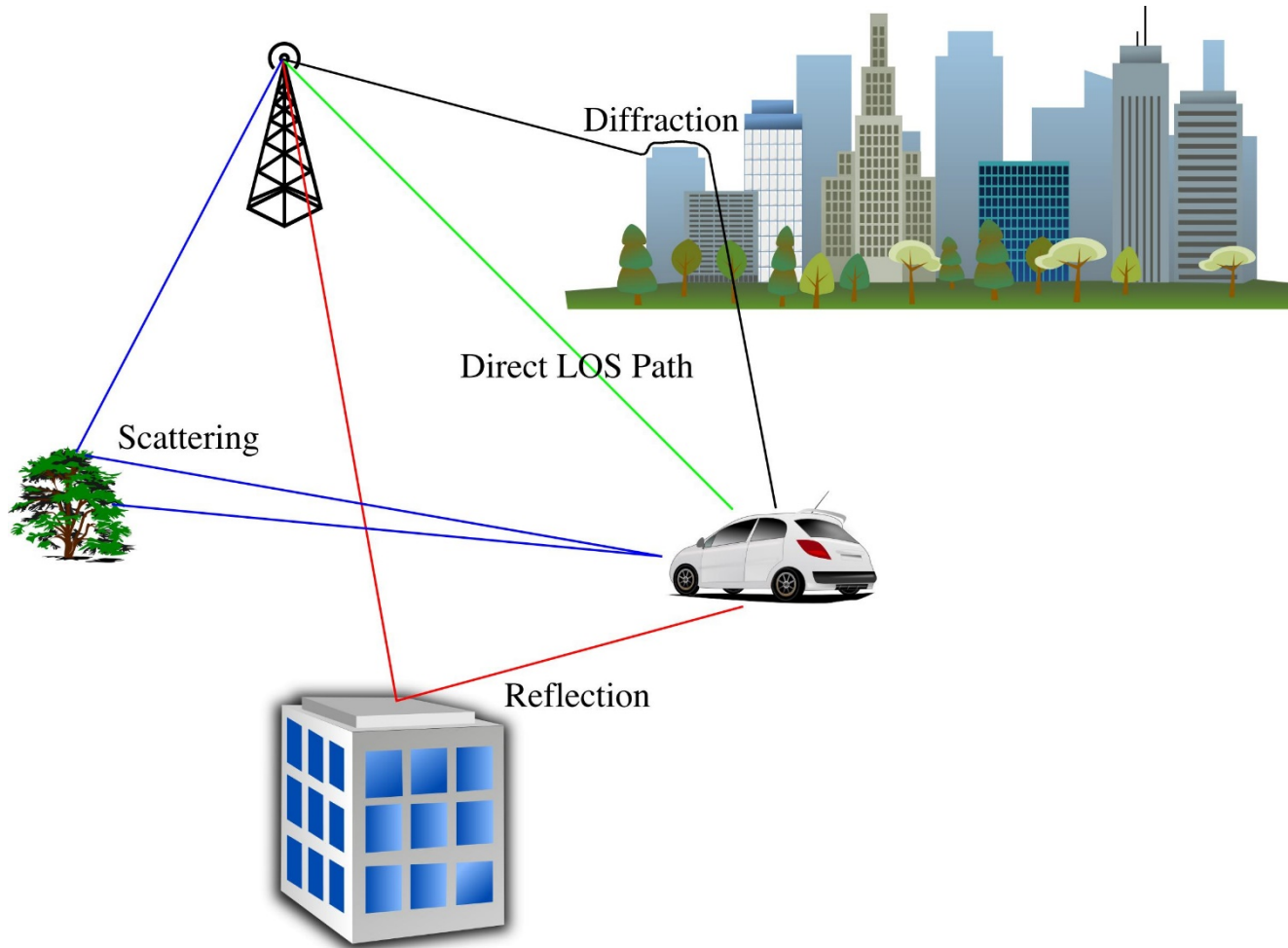
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Multipath Signals in Wireless Communications



How do we model the signal (plus echoes) and remove the echoes?

Discrete-Time System Model

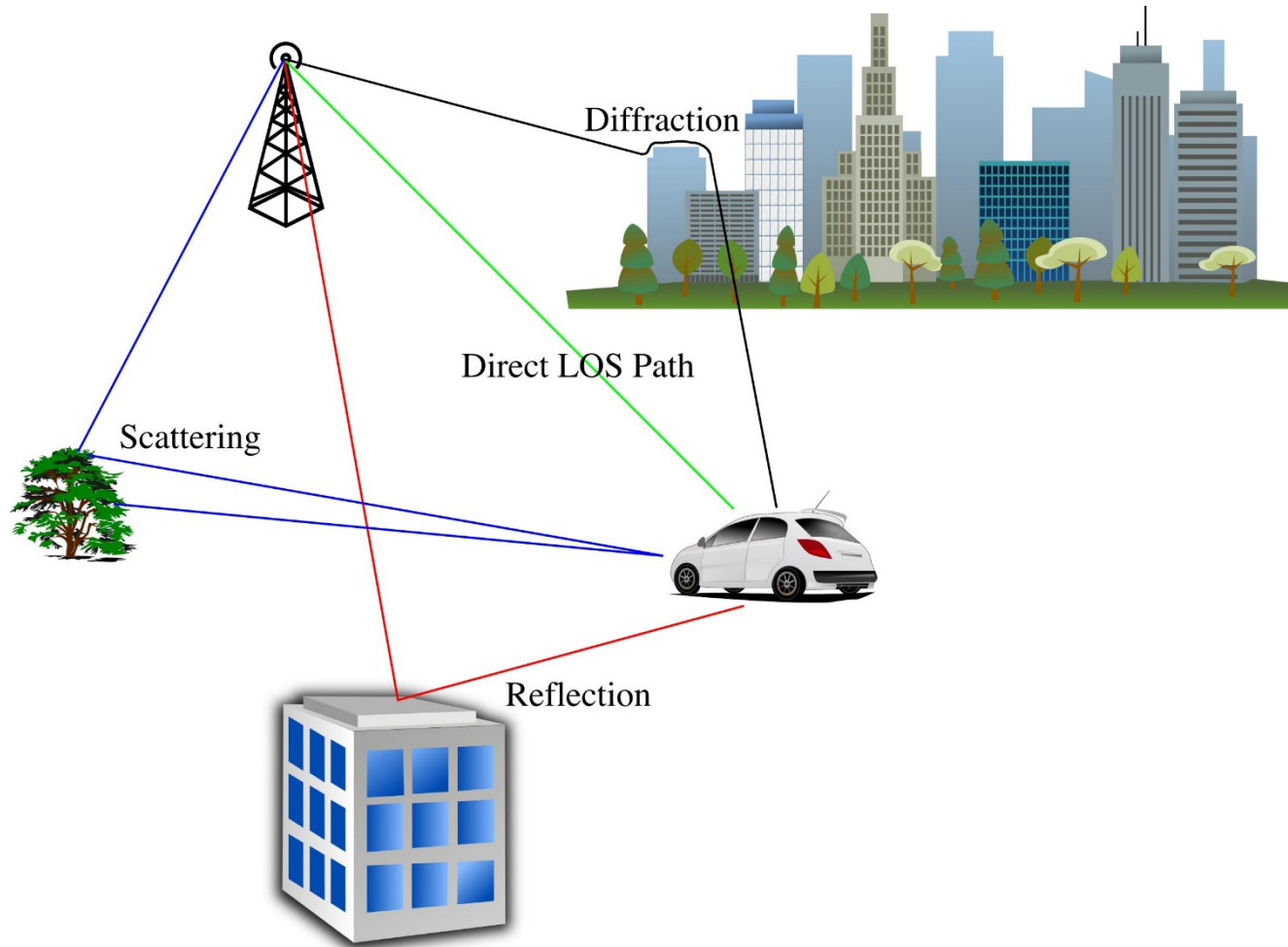


Perhaps only the **green** and **red** components are strong (and the rest are very weak)? If so, our model may look like:

$$y[n] = x[n] - \alpha x[n - N_0]$$

$$\alpha = 0.8, N_0 = 11$$

Using Z Transform (Frequency Domain) to Analyze and “Fix” the System



$$\alpha = 0.8, N_0 = 11$$

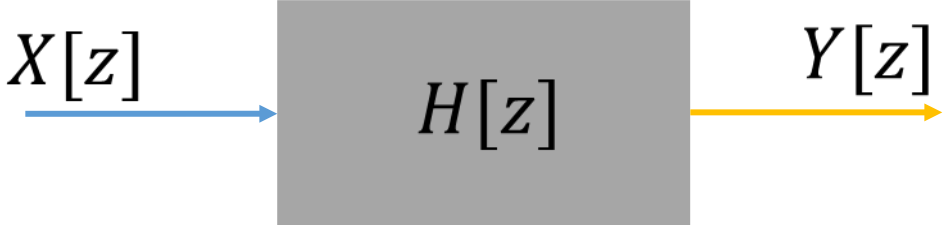
$$y[n] = x[n] - \alpha x[n - N_0]$$

$$Y[z] = X[z] - 0.8Z^{-11}X[z]$$

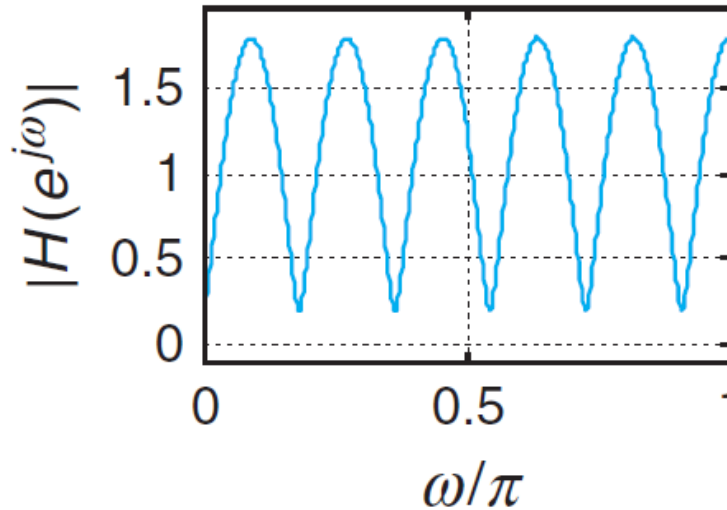
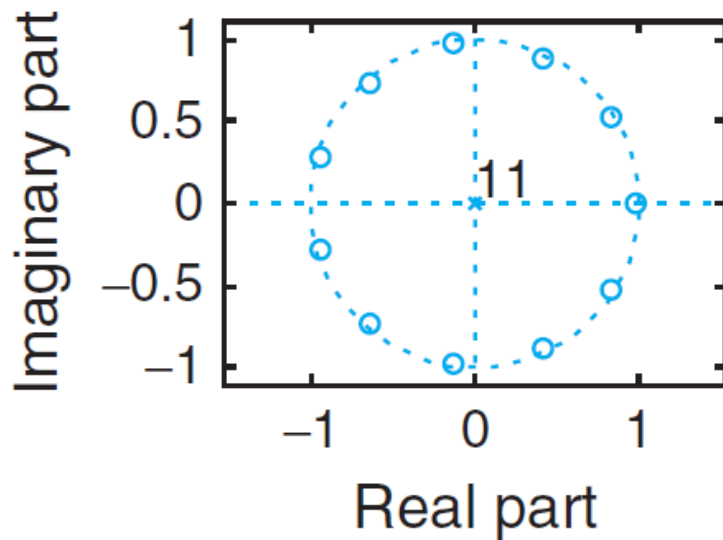
$$Y[z] = (1 - 0.8Z^{-11}) X[z]$$

“System” Transfer Function $H[z]$

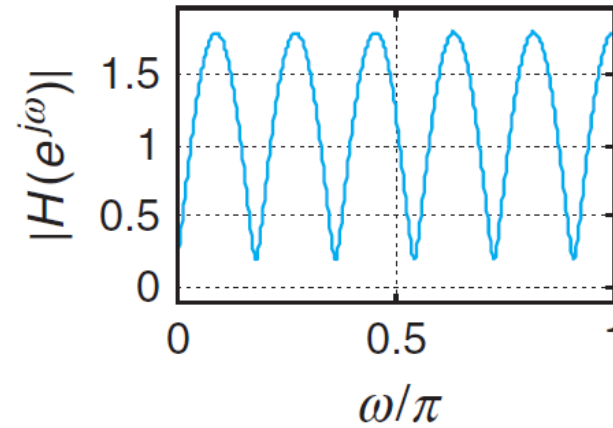
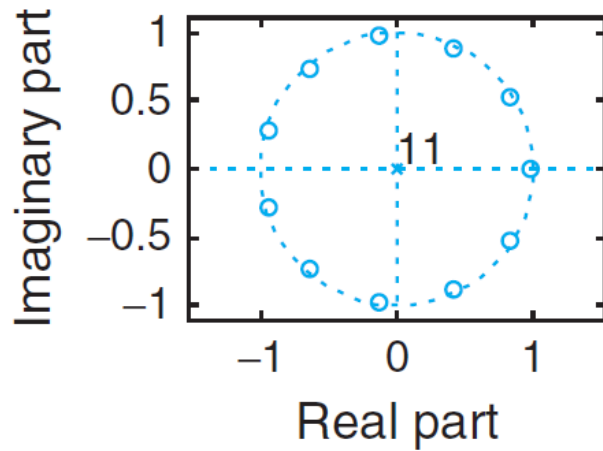
Using Z Transform (Frequency Domain) to Analyze and “Fix” the System

$$H(z) = \frac{Y(z)}{X(z)} = 1 - 0.8z^{-11} = \frac{z^{11} - 0.8}{z^{11}}$$


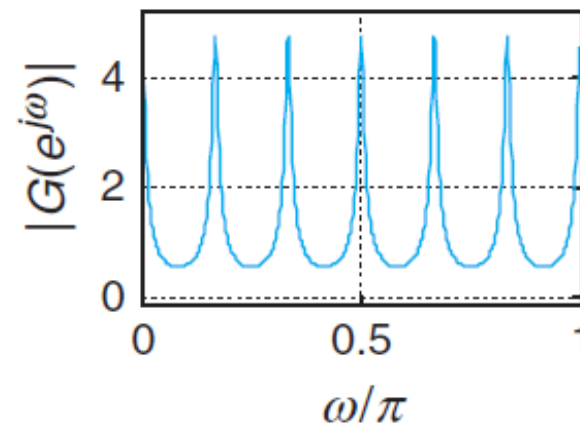
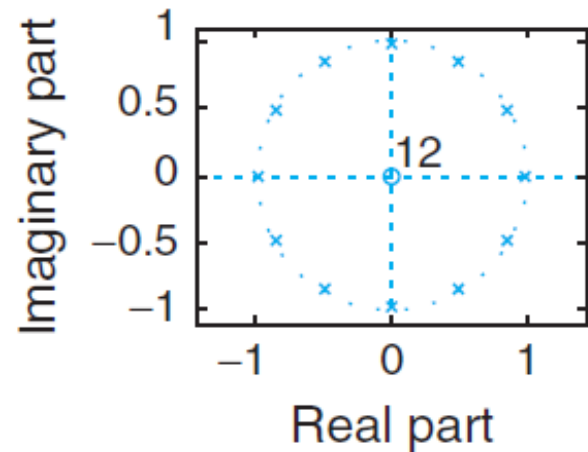
A block diagram showing a system $H[z]$ represented by a gray rectangle. A blue arrow labeled $X[z]$ enters the rectangle from the left, and a yellow arrow labeled $Y[z]$ exits the rectangle to the right.



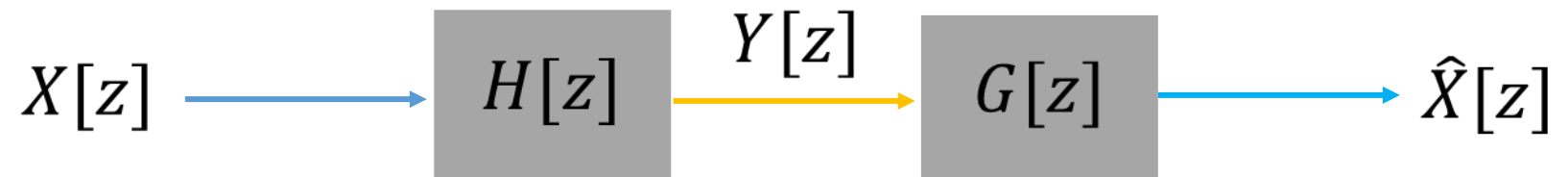
The environment behaves as a “comb” filter. **How may we remove this effect to get original signal back?**

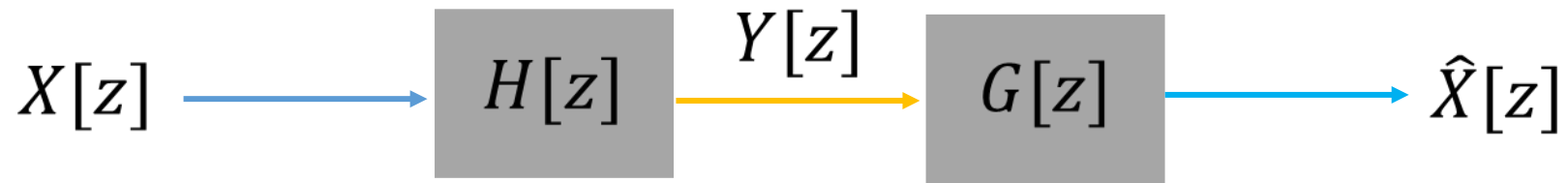


The environment behaves as a “comb” filter. **How may we remove this effect to get original signal back?**



One way is to pass the received signal through a filter that cancels the zeros and poles of the system.

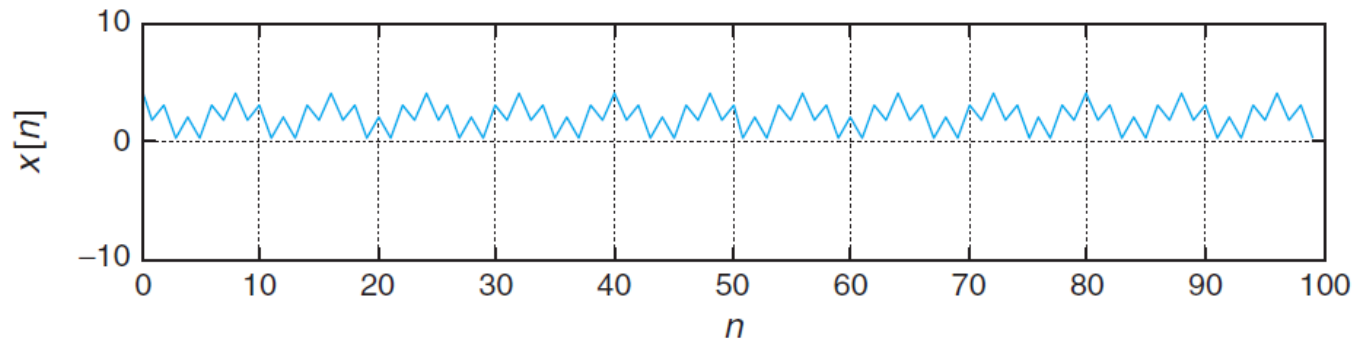
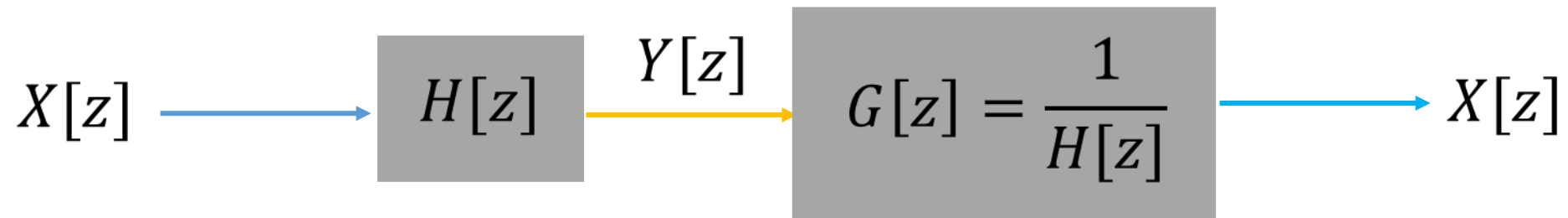




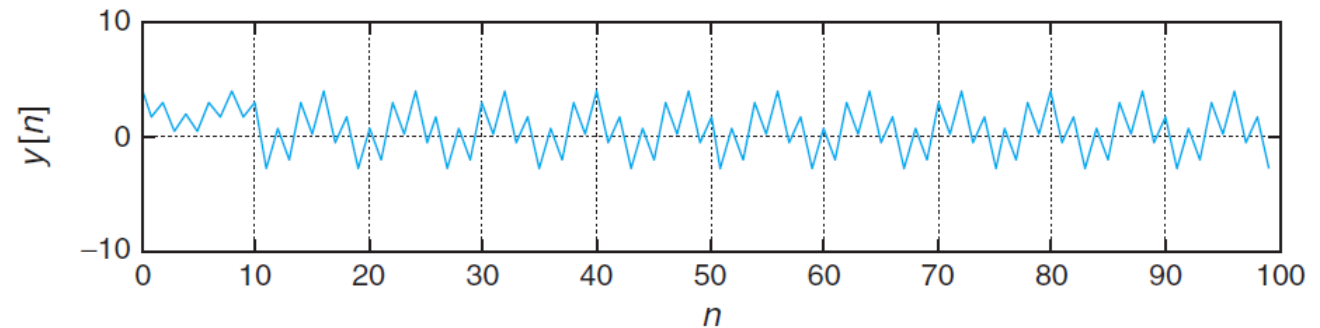
To get $\hat{X}[z] = X[z]$ we must have $H[z]G[z] = 1$, which gives:

$$G(z) = \frac{z^{11}}{z^{11} - 0.8}$$

This filter will remove the effect of multipath and give our original signal back.



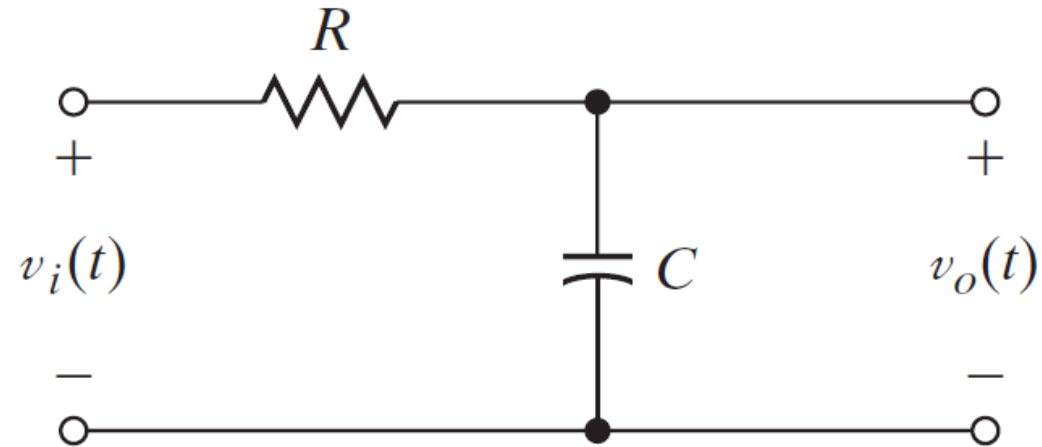
Example signals.



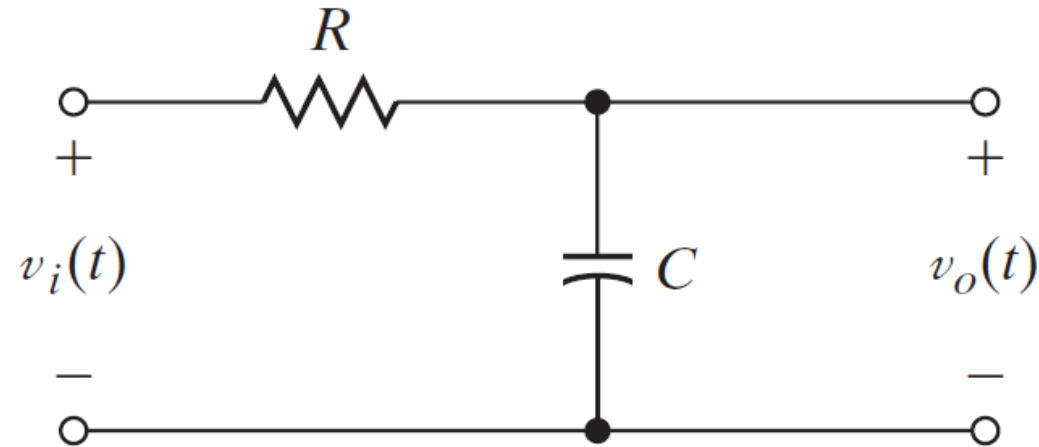
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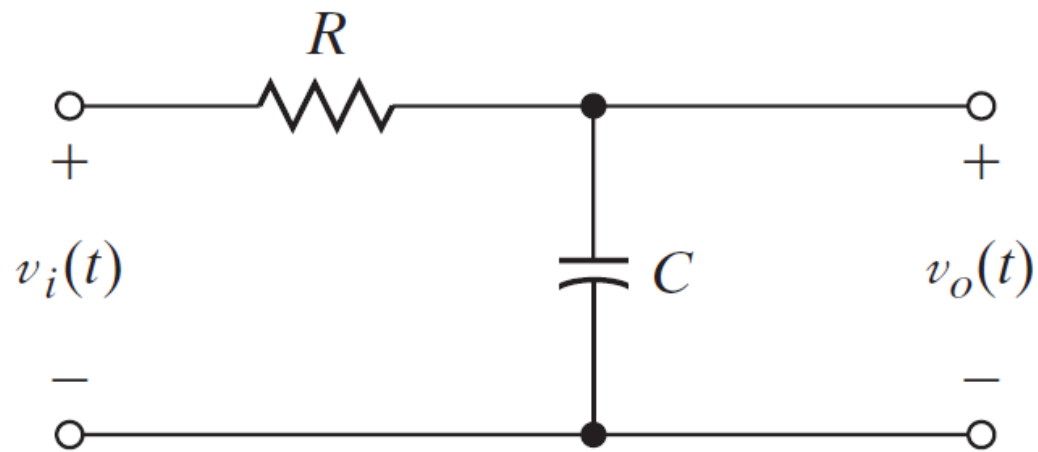
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Example of a Low Pass Filter



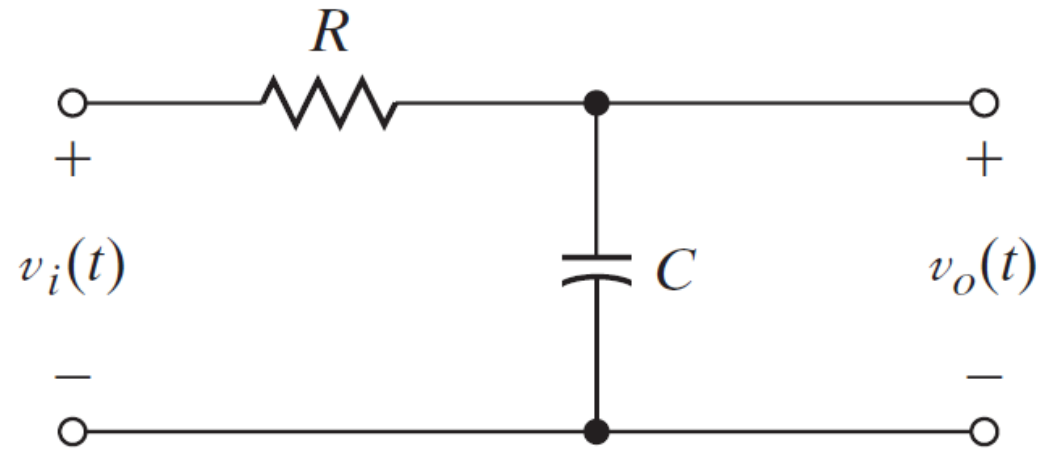
Using Fourier Transform Determine How the Filter's Cutoff Frequency is Linked to R and C





$$v_i(t) = Ri(t) + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau; \quad v_o(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$$

$$V_i(\omega) = RI(\omega) + \frac{1}{j\omega C} I(\omega), V_o(\omega) = \frac{1}{j\omega C} I(\omega)$$



$$v_i(t) = Ri(t) + \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau; \quad v_o(t) = \frac{1}{C} \int_{-\infty}^t i(\tau) d\tau$$

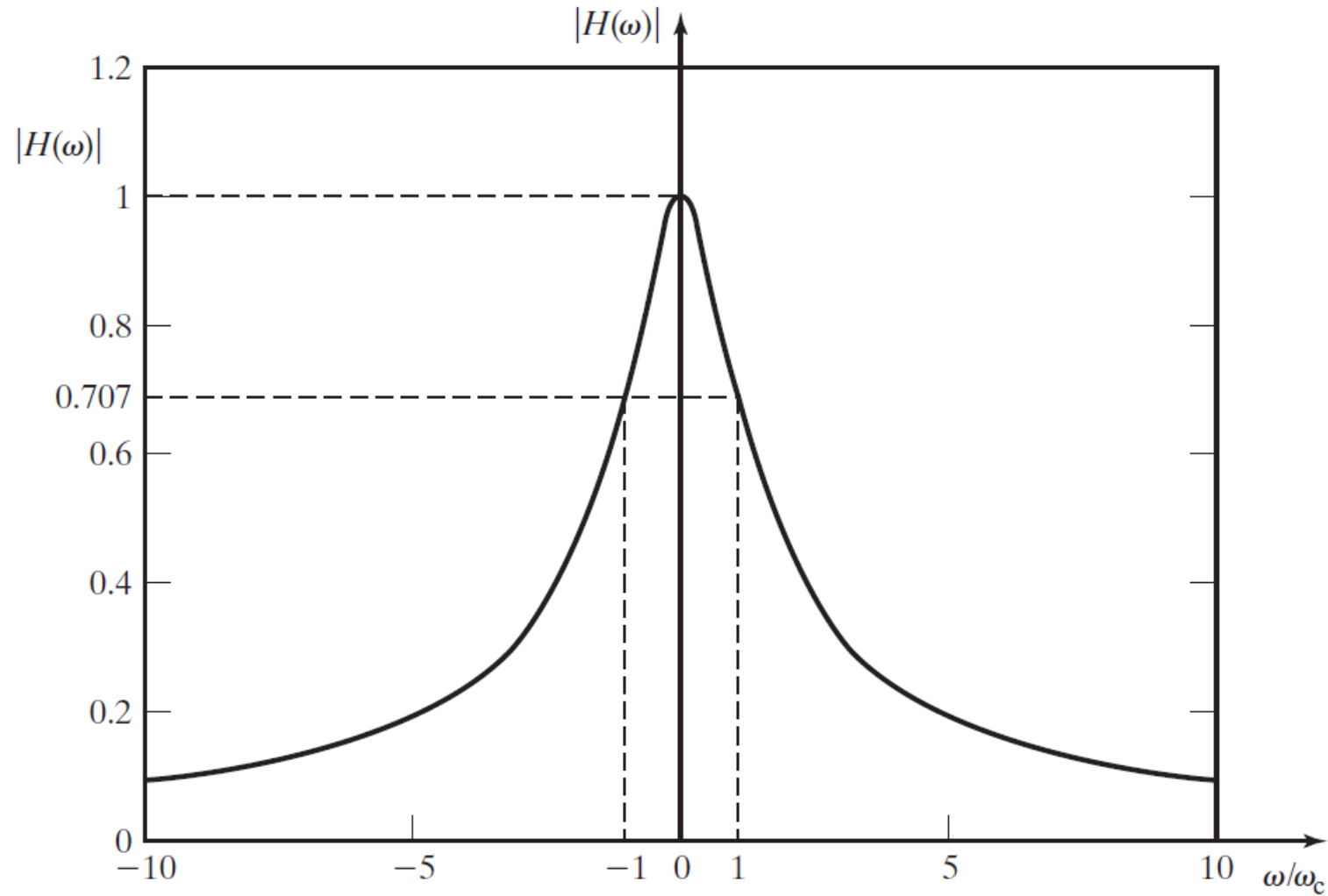
$$V_i(\omega) = RI(\omega) + \frac{1}{j\omega C} I(\omega), V_o(\omega) = \frac{1}{j\omega C} I(\omega)$$

$$H(\omega) = \frac{V_o(\omega)}{V_i(\omega)} = \frac{1}{1 + j\omega RC}.$$

$$|H(\omega)| = \frac{1}{\sqrt{1 + (\omega/\omega_c)^2}}$$

$$\omega_c = \frac{1}{RC}$$

How should you change R and C if you would like to increase (or decrease) the cut-off frequency?



Questions?? Thoughts??

